Abdul Wali Khan University Mardan

BS Final Year Project
Handbook 2025

Department of Computer Science

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Approved for all BS Computer Science Programs

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Preface

This Final Year Project (FYP) Handbook 2025 has been prepared as a comprehensive policy and procedural document to regulate the planning, execution, supervision, and evaluation of Final Year Projects in the Department of Computer Science, Abdul Wali Khan University Mardan. The handbook aims to ensure academic rigor, transparency, fairness, and consistency across all FYP activities.

The handbook is applicable to all students enrolled in the final year of the BS Computer Science program, as well as faculty members involved in supervision and evaluation. Upon approval by the Board of Studies (BOS), the policies outlined herein shall be binding for the academic year 2025.

Acknowledgements

The Department of Computer Science gratefully acknowledges the valuable contributions of its faculty members, the Departmental Final Year Project (FYP) Committee, the Quality Enhancement Cell, and the academic leadership of **Abdul Wali Khan University Mardan.**

Special appreciation is extended to **Dr. Aftab Ahmed Khan, FYP Coordinator**, for his coordination and oversight in the development of this handbook, and to **Prof. Nadeem Iqbal, Chairman, Department of Computer Science**, for his guidance, support, and leadership. Their commitment to academic excellence has played a vital role in ensuring the quality and consistency of this handbook.

1. Introduction

The Final Year Project (FYP) is a mandatory capstone component of the BS Computer Science program at Abdul Wali Khan University Mardan. It represents the culmination of undergraduate academic learning and provides students with an opportunity to integrate theoretical knowledge, practical skills, and professional competencies into a comprehensive and original piece of work.

Through the Final Year Project, students are expected to identify real-world or research-oriented problems, apply appropriate computing methodologies, design and implement effective solutions, and evaluate outcomes critically. The FYP encourages independent thinking, creativity, problem-solving, and the responsible application of modern computing tools and technologies.

The Final Year Project also serves as a platform for students to develop essential professional skills, including project planning, teamwork, technical communication, documentation, and presentation. By engaging in sustained supervision and staged evaluations, students gain experience in managing complex tasks within defined timelines and quality standards.

This handbook has been developed to provide clear guidance on the policies, procedures, roles, evaluation mechanisms, and ethical standards governing the Final Year Project. It is intended to ensure consistency, transparency, and academic rigor across all projects undertaken within the Department of Computer Science.

All students, supervisors, and evaluators are required to adhere to the guidelines outlined in this handbook. Compliance with these provisions ensures that the Final Year Project remains a credible, fair, and academically sound requirement of the BS Computer Science program and contributes to the overall quality and reputation of Abdul Wali Khan University Mardan.

2. Objectives of the Final Year Project

The Final Year Project (FYP) is designed to serve as a comprehensive academic and professional learning experience for students enrolled in the BS Computer Science program at Abdul Wali Khan University Mardan. The primary objective of the FYP is to enable students to apply the theoretical knowledge and practical skills acquired during their undergraduate studies to solve real-world or research-oriented computing problems.

One of the core objectives of the Final Year Project is to develop students' analytical and problem-solving abilities. Through the identification, analysis, and formulation of a problem, students learn to evaluate alternative solutions and select appropriate methodologies, tools, and technologies to address complex computing challenges.

The FYP aims to enhance technical competence by providing hands-on experience in system design, software development, data analysis, experimentation, and implementation. Students are expected to demonstrate proficiency in modern programming languages, development frameworks, computational tools, and emerging technologies relevant to their project domain.

The FYP also focuses on developing essential professional and transferable skills. These include project planning and management, teamwork, ethical decision-making, technical writing, and oral communication. Through regular supervision, documentation, and viva voce examinations, students gain experience in presenting and defending their work in a professional academic setting.

Finally, the Final Year Project aims to instill a strong sense of ethical responsibility and academic integrity. Students are expected to adhere to ethical guidelines, respect intellectual property, ensure originality of work, and consider the societal and professional implications of their projects. Achievement of these objectives ensures that graduates of the BS Computer Science program are well-prepared for professional practice, research, and responsible participation in the computing profession.

3. Governance and Oversight Structure

The governance and oversight of the Final Year Project (FYP) are essential to ensure academic quality, transparency, fairness, and consistency in the planning, supervision, and evaluation of projects. The governance framework for the FYP in the Department of Computer Science, Abdul Wali Khan University Mardan, is designed to provide clear authority, defined responsibilities, and effective monitoring mechanisms throughout the project lifecycle.

The primary academic authority overseeing the Final Year Project is the Departmental Board of Studies (BOS). The BOS is responsible for approving the FYP handbook, evaluation framework, project policies, and any amendments related to the conduct of Final Year Projects. Decisions taken by the BOS provide formal academic legitimacy to the FYP process.

At the departmental level, the Final Year Project Committee functions as the operational body responsible for implementing BOS-approved policies. The committee typically comprises the Chairman of the Department, the FYP Coordinator, and nominated faculty members. It oversees project offerings, supervision allocation, timelines, evaluation schedules, and resolution of academic or administrative issues.

The FYP Coordinator acts as the focal point between the BOS, the Departmental Committee, supervisors, students, and external examiners. The coordinator ensures that policies are implemented uniformly, records are maintained accurately, and all evaluations are conducted in accordance with approved procedures.

Supervisors operate within this governance framework and are accountable for maintaining academic standards, ethical compliance, and continuous supervision of assigned projects. Their role is subject to oversight through documentation, evaluation records, and departmental review.

External examiners provide an independent and impartial assessment of Final Year Projects during the external viva voce. Their involvement strengthens quality assurance, benchmarking, and credibility of the evaluation process.

Through this multi-tier governance and oversight structure, the Department ensures that the Final Year Project process remains robust, accountable, and aligned with university regulations and national quality assurance standards.

4. Roles and Responsibilities

4.1 Role of FYP Coordinator

The Final Year Project (FYP) Coordinator is the primary administrative and academic authority responsible for the overall planning, coordination, and governance of the FYP process within the Department of Computer Science.

The coordinator is responsible for issuing official FYP notifications, announcing project offerings, allocating supervisors, and approving student project groups. Ensuring equitable workload distribution among supervisors is a key responsibility.

The coordinator schedules all internal and external evaluations, coordinates with external examiners, and ensures that evaluation procedures are conducted in accordance with approved policies and timelines.

Record keeping is a critical responsibility of the FYP Coordinator. This includes maintaining official records of supervision meetings, evaluation results, similarity reports, and final grades. The coordinator ensures transparency, consistency, and compliance with departmental and university regulations.

The FYP Coordinator also serves as the primary point of contact for resolving academic and administrative issues related to FYPs. This includes addressing grievances, mediating disputes, reporting matters to the Chairman and Board of Studies (BOS), and recommending policy improvements for continuous quality enhancement.

4.2 Role of Supervisor

The Supervisor plays a central academic and professional role in the successful completion of the Final Year Project (FYP). The Supervisor is responsible for guiding students throughout the entire project lifecycle, from topic refinement to final evaluation. This role requires a balance between academic mentoring, technical oversight, and ethical supervision.

At the initiation stage, the Supervisor assists students in refining the project title, defining the scope, and selecting appropriate methodologies, tools, and technologies. The Supervisor ensures that the project is feasible within the given timeframe and resources, and that it aligns with departmental academic standards.

During the execution phase, the Supervisor conducts regular supervision meetings, monitors progress against agreed milestones, and provides constructive and timely feedback. Supervisors are required to maintain documented supervision records and ensure that students demonstrate continuous progress.

Supervisors are responsible for ensuring originality of work, ethical conduct, and compliance with plagiarism and AI usage policies. They must guide students in proper citation practices and responsible use of emerging technologies.

In the evaluation phase, Supervisors contribute to internal assessments, recommend students for final submission, and support fair and transparent evaluation practices. Supervisors are expected to uphold professionalism, impartiality, and academic integrity at all times.

4.3 Role of Student

The student is the primary stakeholder in the Final Year Project (FYP) and bears the central responsibility for the successful planning, execution, and completion of the project. The FYP requires a high level of commitment, self-discipline, professionalism, and ethical conduct from students throughout the academic year.

At the initial stage, students are responsible for selecting or proposing a suitable project topic, forming groups in accordance with departmental policy, and actively engaging with the allocated supervisor to refine the project scope. Students must ensure that the selected project is feasible, relevant to their discipline, and aligned with academic standards.

During the development phase, students must attend all scheduled supervision meetings, prepare adequately for discussions, and complete assigned tasks and milestones within agreed timelines. Maintaining proper documentation, version control, and progress records is an essential responsibility of the student.

Students are required to ensure originality of work and strict adherence to plagiarism and similarity policies. Proper citation of all sources, responsible use of AI tools, and ethical handling of data are mandatory. Any form of academic dishonesty may result in disciplinary action.

In the evaluation phase, students must prepare for internal and external vivas, submit the final report, code, poster, and similarity report within notified deadlines, and demonstrate a clear understanding of their project. Students are expected to uphold professionalism, respect evaluators, and comply with all departmental regulations.

5. Final Year Project Offering

The Final Year Project (FYP) Offering refers to the formal process through which project titles, domains, and supervision opportunities are announced to final-year students of the BS Computer Science program. This process is designed to ensure academic relevance, feasibility, transparency, and alignment with the educational objectives of Abdul Wali Khan University Mardan.

The Department of Computer Science announces the Final Year Project Offering at the beginning of the final academic year through an official notification issued by the FYP Coordinator. The offering provides students with clear information regarding available project domains, supervision capacity, timelines, and eligibility requirements.

Sources of Project Titles

Final Year Project titles may originate from multiple sources. Faculty members may propose project titles based on their teaching expertise, research interests, funded projects, or industry collaborations. Such projects often provide students with exposure to current research trends and practical problem-solving.

Students may also propose their own project ideas. Student-proposed projects must demonstrate academic relevance, originality, feasibility within the available timeframe, and alignment with the discipline of Computer Science. All student-proposed titles are subject to review and approval by the assigned Supervisor and the Departmental FYP Committee.

Approved Project Domains

Final Year Projects must fall within approved domains of Computer Science. These domains may include, but are not limited to: Software Engineering, Artificial Intelligence, Machine Learning, Data Science, Cybersecurity, Computer Networks, Databases, Web and Mobile Application Development, Internet of Things (IoT), Human–Computer Interaction, and interdisciplinary computing applications.

Projects that fall outside the traditional scope of Computer Science may be considered only if they demonstrate a strong computing component and are approved by the Department.

Group Formation and Project Allocation

Students are required to form project groups in accordance with departmental policy. The size of the group and eligibility criteria are notified at the time of FYP offering. Project allocation is conducted in a transparent manner, taking into account student preferences, project availability, and supervisor workload.

Supervisor Availability and Workload

The number of projects offered by each faculty member is determined by departmental policy and workload considerations. The FYP Coordinator ensures equitable distribution of supervision responsibilities to maintain quality guidance and effective monitoring.

Approval and Finalization of Projects

Once a project title and supervisor are provisionally allocated, students are required to finalize their project through formal proposal submission. Only projects approved through this process are considered official Final Year Projects.

Through a structured and transparent Final Year Project Offering process, the Department ensures that students are provided with academically meaningful, feasible, and well-supervised project opportunities that contribute to the overall quality of the BS Computer Science program.

6. The FYP Process and Timeline

The Final Year Project (FYP) is conducted over two consecutive semesters and follows a structured, time-bound process designed to ensure systematic progress, continuous assessment, and academic rigor. The process emphasizes planning, regular supervision, staged evaluation, and final assessment, enabling students to complete a high-quality project within the prescribed academic timeframe.

Phase 1: Project Announcement and Orientation

At the beginning of the final academic year, the Department of Computer Science formally announces the Final Year Project. An orientation session is conducted by the FYP Coordinator to brief students on project objectives, policies, evaluation criteria, timelines, documentation requirements, and ethical standards. Students are informed about available project domains, supervision capacity, and expectations for successful project completion.

Phase 2: Group Formation and Supervisor Allocation

Following the orientation, students form project groups in accordance with departmental policy. Group formation must be finalized within the notified timeframe. Supervisors are allocated by the FYP Coordinator based on project domain, faculty expertise, and workload considerations. Once allocated, changes in group composition or supervision are discouraged and permitted only under exceptional circumstances.

Phase 3: Proposal Development and Submission

Each project group is required to develop and submit a formal project proposal. The proposal must clearly define the problem statement, objectives, scope, methodology, tools and technologies, expected outcomes, and feasibility analysis. Proposal approval by the assigned Supervisor and FYP Committee is mandatory before proceeding to the implementation phase.

Phase 4: Supervision and Continuous Monitoring

Upon approval of the proposal, students begin the implementation phase under regular supervision. Scheduled supervision meetings are held throughout the semester to monitor progress, resolve technical issues, and ensure adherence to the approved plan. Students are required to demonstrate tangible progress during each meeting, and supervisors maintain documented records of meetings and evaluations.

Phase 5: Internal Viva Evaluations

Two internal viva examinations are conducted during the project lifecycle. The First Internal Viva focuses on problem understanding, literature review, and methodology, while the Second Internal Viva assesses implementation progress, system functionality, and readiness for final submission. Performance in internal vivas contributes significantly to the overall evaluation.

Phase 6: Final Submission

At the conclusion of the development phase, students are required to submit the final FYP report, source code or practical work, project poster, and similarity report within the notified deadline. Submissions must comply with formatting guidelines and academic integrity requirements. Late or incomplete submissions may result in penalties.

Phase 7: External Viva Voce

The External Viva Voce is the final evaluative stage of the FYP. Conducted by an external examiner appointed by the Department, the viva assesses students' conceptual understanding, technical depth, originality, and overall project contribution. Successful completion of the External Viva marks the formal completion of the Final Year Project.

Overall Timeline and Compliance

The FYP timeline is strictly enforced to ensure fairness and academic discipline. Students, supervisors, and evaluators are required to adhere to notified schedules and deadlines. Non-compliance with the approved process or timeline may result in penalties, rescheduling of evaluations, or disciplinary action as per university regulations.

7. Supervision Framework

Effective supervision is a cornerstone of a successful Final Year Project (FYP). The supervision framework outlined in this handbook is designed to ensure continuous academic guidance, systematic monitoring of progress, and consistent application of quality standards across all projects in the Department of Computer Science.

The supervision framework emphasizes structured interaction between students and supervisors, documentation of progress, timely feedback, and accountability of all stakeholders. It also provides mechanisms to support students academically while maintaining fairness, transparency, and academic rigor.

Objectives of the Supervision Framework

The primary objectives of the supervision framework are to:

- Provide sustained academic and technical guidance to students.
- Monitor progress against approved milestones and timelines.
- Identify and address technical or academic challenges at an early stage.
- Ensure originality, ethical conduct, and compliance with academic standards.
- Facilitate fair and evidence-based evaluation.

Supervisor–Student Interaction

Supervision is conducted through regular, scheduled meetings between the supervisor and the student(s). These meetings serve as formal checkpoints to review progress, discuss challenges, and plan subsequent tasks. Students are required to prepare for meetings by documenting completed work, outlining issues, and proposing next steps.

Supervisors provide constructive feedback during meetings and guide students toward feasible solutions. The frequency of meetings is determined by departmental policy, with additional meetings scheduled as needed during critical phases of the project.

Documentation and Meeting Records

Proper documentation is a mandatory component of the supervision framework. Supervisors are required to maintain records of all supervision meetings, including dates, attendance, issues discussed, progress achieved, and action points assigned.

These records serve as evidence of continuous supervision and form the basis for evaluating student engagement, discipline, and progress during the assessment process.

Progress Monitoring and Milestones

The FYP is divided into clearly defined milestones aligned with the approved project proposal and semester-wise timeline. Supervisors monitor progress against these milestones and ensure that students meet expected deliverables within the specified timeframes.

If a student or group fails to demonstrate satisfactory progress, the supervisor may recommend corrective actions, additional meetings, or intervention by the FYP Coordinator.

Academic Integrity and Ethical Oversight

Supervisors play a critical role in ensuring academic integrity throughout the project lifecycle. This includes guiding students on proper citation practices, responsible use of AI tools, and ethical handling of data.

Any suspected cases of plagiarism, misconduct, or unethical behavior must be reported to the FYP Coordinator for further action in accordance with university regulations.

Role of the FYP Coordinator in Supervision

The FYP Coordinator oversees the implementation of the supervision framework at the departmental level. The coordinator ensures consistency in supervision practices, reviews supervision records, and addresses issues related to supervision quality or workload imbalance.

Escalation and Support Mechanism

In cases where supervision-related issues arise, students may approach the FYP Coordinator through formal channels. The escalation mechanism ensures that academic concerns are addressed fairly while maintaining respect for the supervisor–student relationship.

Linkage with Evaluation

Supervision records and documented progress form a key component of the evaluation framework. Attendance, preparedness, responsiveness to feedback, and milestone completion directly influence marks allocated for supervision and continuous assessment.

The supervision framework thus ensures that the Final Year Project is not only assessed on final output but also on the quality of the academic process followed throughout the year.

8. Evaluation Framework

The Evaluation Framework for the Final Year Project (FYP) is designed to ensure fairness, transparency, academic rigor, and consistency in the assessment of student performance. The framework emphasizes both the project development process and the final project outcomes, ensuring that students are evaluated holistically rather than solely on the final deliverable.

The FYP carries a total of 200 marks, distributed across continuous assessment, internal evaluations, and final evaluation. This structured distribution allows for regular monitoring of progress, timely feedback, and objective assessment at different stages of the project lifecycle.

Principles of Evaluation

The evaluation framework is guided by the following principles:
• Transparency: Clear communication of evaluation criteria to all stakeholders.
• Consistency: Uniform application of approved rubrics across all projects.
• Objectivity: Evidence-based assessment supported by documentation and records.
• Continuous Assessment: Emphasis on sustained effort throughout the project duration.
• Academic Integrity: Zero tolerance for plagiarism and unethical practices.

Components of Evaluation

The FYP evaluation comprises multiple components, each addressing a specific aspect of project development and learning outcomes. These components collectively contribute to the final grade and ensure balanced assessment.

Continuous Supervision (Monthly Meetings): Marks are allocated based on attendance, preparedness, progress demonstration, and professional conduct during supervision meetings.

Proposal Evaluation: Assesses clarity of problem statement, feasibility, methodology, and alignment with academic objectives.

Internal Viva Evaluations: Two internal vivas evaluate conceptual understanding, literature review, methodology, implementation progress, and readiness for final submission.

External Viva Voce: Conducted by an external examiner, this component assesses overall project quality, technical depth, originality, and the student's ability to defend the work.

Project Report and Practical Work: Evaluates documentation quality, analysis, implementation, innovation, and reproducibility.

Similarity Report: Ensures compliance with academic integrity standards.

Marks Distribution and Weightage

The approved marks distribution for the FYP is as follows:

- Monthly Meetings (Supervision): 50 marks
- Proposal Submission: 10 marks
- First Internal Viva: 25 marks
- Second Internal Viva: 25 marks
- External Viva Voce: 50 marks
- Thesis / Report: 15 marks
- Code / Practical Work: 15 marks
- Similarity Report: 10 marks

Total: 200 marks

Use of Evaluation Rubrics

Standardized evaluation rubrics are used for each component to ensure consistency and fairness. Rubrics define performance levels, descriptors, and marks allocation criteria, enabling evaluators to justify awarded marks transparently.

Documentation and Evidence

All evaluation decisions must be supported by documentary evidence, including supervision logs, viva evaluation forms, similarity reports, and final submission records. Proper documentation ensures accountability and facilitates review or audit if required.

Role of Internal and External Examiners

Internal examiners assess continuous progress and technical development, while the external examiner provides an independent and unbiased evaluation of the final project. The combined assessment ensures balance between internal familiarity and external objectivity.

Handling of Discrepancies and Appeals

In case of significant discrepancies or student appeals, the matter may be reviewed by the FYP Coordinator or Departmental Committee in accordance with university regulations. All decisions are documented and communicated formally.

Linkage with Learning Outcomes

The evaluation framework is aligned with the Program Learning Outcomes (PLOs) of the BS Computer Science program. Assessment components collectively measure technical competence, problem-solving ability, communication skills, ethical awareness, and professional readiness.

Through this comprehensive evaluation framework, the Department ensures that the Final Year Project serves as a credible and rigorous measure of student achievement and preparedness for future professional endeavors.

Marks Distribution Overview

The following pie chart illustrates the approved distribution of marks for the Final Year Project (Total = 200 marks). It visually represents the weightage assigned to continuous assessment, internal evaluations, external viva, documentation, practical work, and academic integrity. This distribution ensures a balanced evaluation of both process and final outcomes.

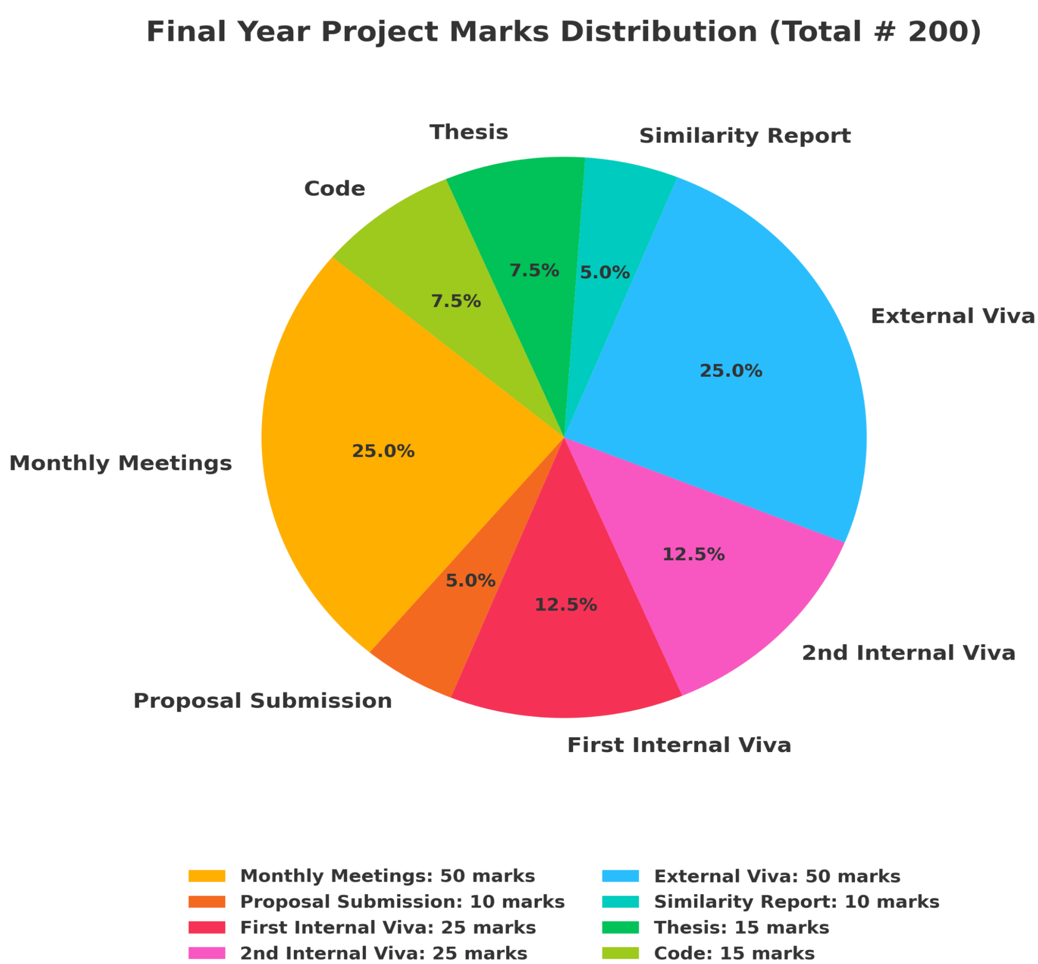

Figure: Final Year Project Marks Distribution (Total = 200 Marks)

9. Viva Voce Examinations

Viva Voce examinations constitute a critical component of the Final Year Project (FYP) evaluation process. They are designed to assess a student's depth of understanding, technical competence, originality of work, and ability to articulate and defend the project in a professional academic setting.

The Viva Voce process ensures that the Final Year Project is not only evaluated based on written documentation and implementation but also on the student's conceptual clarity, analytical reasoning, and communication skills.

Types of Viva Voce Examinations

The FYP evaluation process includes the following Viva Voce examinations:

- **First Internal Viva**
- **Second Internal Viva**
- **External Viva Voce**

Each viva serves a distinct purpose and is conducted at specific stages of the project lifecycle.

First Internal Viva

The First Internal Viva is conducted during the early phase of the project. Its primary objective is to evaluate the student's understanding of the problem domain, clarity of objectives, literature review, and proposed methodology.

During this viva, students are expected to demonstrate a clear understanding of the problem statement, justify their project scope, and explain the chosen methodology. Feedback provided during this stage is formative and aimed at improving project direction and quality.

Second Internal Viva

The Second Internal Viva is conducted at the mid-stage of the project and focuses on implementation progress. Students are evaluated on system design, development progress, preliminary results, and challenges encountered.

This viva assesses whether the project is progressing according to the approved plan and whether students are adequately prepared for final submission. Recommendations made during this stage are critical for successful completion.

External Viva Voce

The External Viva Voce represents the final and most comprehensive evaluation of the Final Year Project. It is conducted by an external examiner appointed by the Department to ensure impartial and independent assessment.

During the External Viva, students must demonstrate complete ownership of the project, explain design and implementation decisions, interpret results, and respond effectively to technical questions. The External Viva significantly contributes to the final grade.

Viva Panel Composition

Internal Viva panels typically consist of departmental faculty members, including the supervisor. The External Viva panel includes the external examiner, internal examiners, and the FYP Coordinator or nominee.

Assessment Criteria

Viva Voce examinations assess multiple dimensions of student performance, including:

- Conceptual understanding
- Technical depth and implementation
- Originality and innovation
- Problem-solving ability
- Communication and presentation skills
- Response to questions and critical feedback

Conduct and Professionalism

Students are expected to maintain professional conduct during all Viva Voce examinations. This includes punctuality, appropriate dress, respectful communication, and adherence to departmental guidelines.

Documentation and Record Keeping

Viva evaluation forms must be completed by examiners and retained as official records. These records contribute to transparency, accountability, and future reference.

Handling Absence and Re-Viva

Absence from a scheduled Viva Voce without valid justification may result in zero marks or rescheduling at the discretion of the Department. Re-viva may be conducted only under exceptional circumstances and in accordance with university regulations.

Integration with Overall Evaluation

Viva Voce performance is integrated into the overall FYP evaluation framework. Marks awarded during vivas reflect both the quality of the project and the student's ability to defend it academically.

Through a structured and transparent Viva Voce examination process, the Department ensures that the Final Year Project serves as a credible measure of student competence and readiness for professional or academic advancement.

10. Guidelines for the Preparation of FYP Poster

The Final Year Project (FYP) poster is an essential academic artifact that provides a concise visual summary of the project. It is used during viva voce examinations, project exhibitions, and departmental reviews to communicate the core idea, methodology, implementation, and outcomes of the project in a clear and professional manner.

The purpose of the FYP poster is to enable evaluators and viewers to quickly understand the scope and contribution of the project without referring to the complete written report. Students are therefore expected to prepare the poster with clarity, accuracy, and academic professionalism.

Poster Size and Format

The standard size of the FYP poster shall be A0 unless otherwise notified by the Department. The poster may be oriented in either portrait or landscape format. All posters must be printed clearly and presented in a professional layout.

Mandatory Poster Components

Each FYP poster must include the following components

- Project Title
- Student Name(s) and Registration Number(s)
- Supervisor Name
- Department and University Name
- Abstract or Project Overview
- Objectives
- Methodology
- System Architecture / Design
- Implementation Highlights
- Results and Key Findings
- Conclusion and Future Work

Content Guidelines

Poster content should be concise and focused. Long paragraphs should be avoided, and information should be presented using bullet points, diagrams, flowcharts, tables, and figures wherever possible. The text should communicate the essence of the project without excessive technical detail.

Figures and diagrams must be clearly labeled and readable from a reasonable viewing distance. Any borrowed images or content must be properly cited to maintain academic integrity.

Font and Readability

Appropriate font sizes must be used to ensure readability. As a general guideline, the project title should be prominently displayed, section headings should be clearly visible, and body text should be readable from a distance of at least one meter.

Visual Design and Layout

The poster layout should follow a logical flow, typically from left to right and top to bottom. Adequate spacing between sections should be maintained to avoid clutter. Consistent color schemes and alignment should be used to enhance visual appeal and professionalism.

Use of Graphics and Charts

Graphics, charts, and system diagrams are strongly encouraged, as they improve comprehension and engagement. Graphs and charts should be accurately labeled, with units and legends clearly indicated.

Ethical and Academic Considerations

The poster must represent original work. Any form of plagiarism or misrepresentation of results is strictly prohibited. Students must ensure that data presented in the poster is consistent with the project report and implementation.

Poster Submission and Presentation

Students are required to submit the FYP poster by the notified deadline. During viva examinations, students must be able to explain each section of the poster confidently and respond to questions raised by examiners.

Evaluation of FYP Poster

The FYP poster is evaluated based on clarity of presentation, organization of content, visual quality, accuracy of information, and the student’s ability to explain the poster during the viva voce. Poster evaluation contributes to the overall assessment of the Final Year Project.

By adhering to these guidelines, students can ensure that their FYP poster effectively communicates their work and reflects the academic standards of the Department of Computer Science, Abdul Wali Khan University Mardan.

11. Guidelines for FYP Report Format

The Final Year Project (FYP) report is the primary academic document that records the complete process, methodology, implementation, and outcomes of the project. It reflects the student's ability to conduct systematic inquiry, apply technical knowledge, analyze results, and communicate findings in a clear, structured, and professional manner. The FYP report must therefore adhere strictly to standardized academic formatting guidelines prescribed by the Department of Computer Science.

These guidelines are intended to ensure uniformity, readability, and academic integrity across all Final Year Project submissions. Non-compliance with the prescribed format may result in deductions or the requirement to resubmit the report.

General Formatting Requirements

The FYP report must be prepared using a standard word-processing tool. The prescribed font is Times New Roman, with a font size of 12 points for the main text. Line spacing must be set to 1.5 throughout the document, with text fully justified. Margins should be set uniformly on all sides to ensure adequate space for binding and annotations.

All pages must be numbered consecutively. Preliminary pages may use Roman numerals, while the main body of the report should use Arabic numerals. Consistent formatting must be maintained throughout the document.

Structure and Organization

The FYP report must be organized in a logical and coherent manner. Headings and subheadings should follow a clear hierarchical structure. Chapter titles must be centered and clearly distinguished from sub-sections. Consistency in heading styles is mandatory.

Language and Writing Style

The report must be written in clear, formal academic English. Informal language, colloquial expressions, and unsupported claims should be avoided. Technical terms must be used accurately, and abbreviations should be defined upon first use.

Students are encouraged to write in a concise and objective manner, ensuring logical flow between sections and coherence within chapters.

Tables, Figures, and Illustrations

Tables, figures, diagrams, and illustrations must be properly numbered and titled. Each table or figure should be referenced in the text and placed as close as possible to its first mention. Captions must be clear and descriptive.

All graphical elements must be of high quality and readable. Sources of borrowed figures or data must be properly acknowledged to maintain academic integrity.

Equations and Algorithms

Mathematical equations, algorithms, and pseudocode must be clearly presented and properly numbered. Symbols and variables should be defined where necessary. Algorithms may be presented using standard formats or structured pseudocode for clarity.

Referencing and Citation Style

All sources consulted during the project must be properly cited using an approved referencing style. In-text citations and the reference list must be consistent. Plagiarism, including improper paraphrasing or missing citations, is strictly prohibited.

Consistency and Quality Control

Before submission, students must ensure consistency in formatting, spelling, grammar, and presentation. Proofreading and peer review are strongly recommended. The final submitted report should represent a polished and professional academic document.

Submission Requirements

The final FYP report must be submitted in both hard copy and electronic form as notified by the Department. The electronic version must be identical to the printed copy. Late submissions or reports not conforming to the prescribed format may be subject to penalties.

By adhering to these guidelines, students ensure that their Final Year Project report meets the academic standards of the Department of Computer Science and effectively communicates the depth and quality of their work.

12. FYP Report Contents (Chapter-wise)

The Final Year Project (FYP) report is expected to present the complete academic and technical journey of the project in a structured and logical manner. Each chapter of the report serves a specific purpose and contributes to the overall coherence, rigor, and academic quality of the document. Students must ensure that all chapters are written clearly, systematically, and in accordance with approved academic standards.

Title Page

The title page must include the project title, student name(s), registration number(s), degree program, department, university name, supervisor name, and year of submission. The title should be concise, descriptive, and reflective of the project scope.

Abstract

The abstract provides a brief summary of the entire project, typically within one page. It should concisely describe the problem addressed, objectives, methodology, key results, and conclusions. The abstract should be self-contained and written in clear academic language.

Chapter 1: Introduction

The introduction chapter outlines the background of the study, problem statement, objectives, scope, and significance of the project. It should clearly define the motivation for the project and provide context for the work undertaken.

Chapter 2: Literature Review

This chapter presents a critical review of existing research, technologies, or systems relevant to the project. Students should analyze and compare prior work, identify gaps, and justify how the current project addresses these gaps.

Chapter 3: Methodology

The methodology chapter describes the overall approach adopted to solve the problem. It should explain system models, algorithms, research design, tools, technologies, datasets, and procedures used. The methodology must be detailed enough to allow replication of the work.

Chapter 4: System Design / Experimental Setup

This chapter explains the system architecture, design models, flow diagrams, or experimental setup. Design decisions should be justified, and diagrams should be clearly labeled and explained.

Chapter 5: Implementation

The implementation chapter details the development process, including modules, components, coding practices, and integration steps. Screenshots, code snippets, or implementation details may be included where appropriate.

Chapter 6: Results and Discussion

This chapter presents experimental results, system outputs, performance evaluation, and analysis. Results should be interpreted critically, and comparisons with existing approaches may be included where relevant.

Chapter 7: Conclusion and Future Work

The conclusion summarizes the project outcomes, achievements, and limitations. This chapter should also outline possible future enhancements or research directions.

References

All sources cited in the report must be listed in the references section using the approved referencing style. Consistency and accuracy in citations are mandatory.

Appendices

Appendices may include supplementary materials such as extended data, questionnaires, code samples, or additional documentation that supports the main text but is not essential to include within the chapters.

By following these chapter-wise guidelines, students ensure that their FYP report is comprehensive, coherent, and meets the academic expectations of the Department of Computer Science, Abdul Wali Khan University Mardan.

13. The Proposal Template

This proposal template is provided to guide students in preparing a clear, structured, and academically sound Final Year Project proposal. It outlines the required sections and formatting to ensure consistency and compliance with departmental and university guidelines. Students are advised to complete each section carefully, in consultation with their supervisor, before submission for approval. Adherence to this template will facilitate timely review and fair evaluation of project proposals.

FINAL YEAR PROJECT PROPOSAL

Proposed Title of the Project

BS Computer Science
Department of Computer Science
Abdul Wali Khan University Mardan

Student Name: ________________________
Registration No: _____________________

Supervisor: __________________________

Session / Year: 2025

- **Introduction**

Provide background of the problem domain and context of the proposed project.

- **Problem Statement**

Clearly define the problem that the project intends to address.

- **Objectives of the Project**

List the main objectives and expected outcomes of the proposed project.

- **Scope of the Project**

Describe the boundaries, limitations, and coverage of the project.

- **Literature Review (Brief)**

Summarize relevant existing work, technologies, or research related to the project.

- **Proposed Methodology**

Explain the approach, tools, techniques, models, algorithms, or frameworks to be used.

- **Tools and Technologies**

List programming languages, frameworks, software tools, and platforms to be used.

- **Expected Outcomes**

Describe the anticipated results, deliverables, or contributions of the project.

- **Project Timeline**

Provide a tentative timeline or Gantt chart for project activities.

- **Ethical Considerations**

Describe ethical, legal, and academic integrity considerations related to the project.

- **References**

List references using APA / IEEE style.

- **Approval**

The undersigned recommend approval of this project proposal.

Supervisor Signature: ____________________ Date: ___________

FYP Coordinator Signature: _______________ Date: ___________

14. The Thesis Template

This thesis template has been designed to provide a standardized structure and formatting framework for the preparation of the BS Final Year Project thesis. It ensures uniformity, clarity, and academic consistency across all project submissions within the Department of Computer Science. By following this template, students can effectively organize their work, present their research and development activities systematically, and comply with departmental and university requirements.

TITLE OF THE FINAL YEAR PROJECT

A Thesis Submitted in Partial Fulfillment of the Requirements
for the Degree of

BS Computer Science

Department of Computer Science
Abdul Wali Khan University Mardan

By

Student Name (Registration No.)

Supervisor:
Dr. ___________________

Year: 2025

Declaration

I hereby declare that this thesis is my own work and has not been submitted previously for any degree. All sources of information used have been properly acknowledged.

Student Name
Signature:
Date:

Certificate of Approval

This is to certify that the Final Year Project titled "________________________" has been carried out by ________________________ under my supervision and is approved for submission in partial fulfillment of the requirements for the BS Computer Science degree.

Supervisor:
Signature:
Date:

Abstract

The abstract should summarize the project problem, objectives, methodology, results, and conclusions in approximately 250–300 words.

Table of Contents

Generate automatically using MS Word.

Chapter 1: Introduction

Write content here...

Chapter 2: Literature Review

Write content here...

Chapter 3: Methodology

Write content here...

Chapter 4: System Design / Experimental Setup

Write content here...

Chapter 5: Implementation

Write content here...

Chapter 6: Results and Discussion

Write content here...

Chapter 7: Conclusion and Future Work

Write content here...

References

Follow APA / IEEE referencing style.

15. The Viva Forms

The following evaluation forms have been designed to ensure a structured, transparent, and consistent assessment of Final Year Projects during internal and external viva voce examinations. All examiners are required to complete the relevant form at the time of evaluation, and the completed forms shall form part of the official FYP assessment record.

- **Internal Viva Evaluation Form**

Student Name: ______________________________

Registration No: ____________________________

Project Title: ____

Supervisor: _________________________________

Date of Viva: ________________________________

Evaluation Criteria:

Criteria	Marks Obtained	Maximum Marks
Problem Understanding & Objectives		5
Literature Review & Background Knowledge		5
Methodology / Design Approach		5
Progress & Implementation Status		5
Presentation & Communication Skills		5

Total Marks: _______ / 25

Comments:

____

____

Examiner Name & Signature: _________________

- **External Viva Voce Evaluation Form**

Student Name: ________________________________

Registration No(s): ____________________________

Project Title: ____

External Examiner: ____________________________

Date of Viva: _________________________________

Evaluation Criteria:

Criteria	**Marks Obtained**	**Maximum Marks**
Conceptual Understanding & Defense		10
Technical Depth & Implementation		10
Originality & Innovation		10
Results, Analysis & Discussion		10
Presentation & Response to Questions		10

Total Marks: ______ / 50

Overall Remarks:

____

____

External Examiner Signature: ______________

16. Similarity and Plagiarism Policy

Academic integrity is a fundamental principle of higher education and a core value of Abdul Wali Khan University Mardan. The Final Year Project (FYP) is expected to reflect original intellectual effort, ethical research practices, and proper acknowledgement of all sources. This Similarity and Plagiarism Policy outlines the standards, procedures, and consequences related to academic dishonesty in Final Year Projects.

All students, supervisors, and evaluators involved in the FYP process are required to adhere strictly to this policy. Failure to comply may result in penalties as prescribed by departmental and university regulations.

Definition of Plagiarism

Plagiarism refers to the use of another person’s ideas, words, data, code, designs, or results without proper acknowledgement. This includes, but is not limited to, direct copying, improper paraphrasing, self-plagiarism, unauthorized collaboration, and the use of purchased or ghost-written material.

Similarity Checking

All Final Year Project reports are subject to similarity checking using plagiarism detection software approved by the University. The similarity report provides a percentage indicating textual overlap with existing sources. Similarity checking is conducted at the time of final submission and may also be carried out at intermediate stages if deemed necessary by the Supervisor or FYP Coordinator.

Acceptable Similarity Threshold

The acceptable similarity threshold for Final Year Projects shall be as per prevailing university policy. Similarity arising from properly cited references, standard definitions, or commonly used technical terms may be considered acceptable at the discretion of the evaluators.

Student Responsibilities

Students are responsible for ensuring originality of their work. Proper citation and referencing of all sources must be followed throughout the project report. Students must ensure that similarity reports are generated and submitted as part of the final FYP documentation.

The use of Artificial Intelligence (AI) tools for assistance must be disclosed where applicable. Core intellectual contributions, analysis, and implementation must remain the student’s own work.

Supervisor Responsibilities

Supervisors play a critical role in promoting academic integrity. They are responsible for guiding students on proper citation practices, monitoring originality during supervision, and reviewing similarity reports before recommending projects for final submission.

Handling of Similarity and Plagiarism Cases

If the similarity index exceeds the acceptable threshold, the project may be returned to the student for revision and resubmission. In cases of severe or intentional plagiarism, the matter shall be referred to the Departmental Committee for disciplinary action in accordance with university rules.

Penalties

Penalties for plagiarism may include reduction of marks, requirement to revise the report, failure in the project, or further disciplinary action as per university statutes. The severity of the penalty depends on the extent and nature of the violation.

Documentation and Record Keeping

All similarity reports and related correspondence must be retained as official records by the Department. These records ensure transparency and support audit, review, or appeal processes.

Appeal Mechanism

Students may submit a formal appeal against a plagiarism-related decision through the prescribed departmental channels. Appeals are reviewed in accordance with university regulations, and decisions of the competent authority shall be final.

By enforcing this Similarity and Plagiarism Policy, the Department of Computer Science ensures that the Final Year Project remains a credible, ethical, and academically rigorous component of the BS Computer Science program.

17. Ethical Guidelines

Ethical conduct is a fundamental requirement of academic work and professional practice. The Final Year Project (FYP) must be conducted in accordance with high ethical standards to ensure integrity, credibility, and social responsibility. These Ethical Guidelines outline the principles and expectations governing ethical behavior throughout the FYP lifecycle.

All students, supervisors, and evaluators are required to comply with these guidelines. Any violation of ethical standards may result in disciplinary action as per departmental and university regulations.

Principles of Ethical Conduct

The ethical conduct of the FYP is guided by the following principles:

- Honesty and integrity in all academic activities.
- Respect for intellectual property and proper acknowledgment of sources.
- Responsibility and accountability in research and development activities.
- Transparency in data collection, analysis, and reporting.
- Respect for individuals, society, and the environment.

Originality and Academic Honesty

Students must ensure that all aspects of the FYP represent original work. Plagiarism, fabrication of data, falsification of results, or misrepresentation of contributions are strictly prohibited. Proper citation and referencing of all sources are mandatory.

Responsible Use of Technology and AI Tools

The use of software tools, online resources, and Artificial Intelligence (AI) systems must be responsible and ethical. AI tools may be used for assistance in areas such as proofreading, idea generation, or coding support; however, the core intellectual contribution, analysis, and implementation must be the student's own work.

Any use of AI tools that significantly influences the project must be disclosed in the report where applicable.

Data Privacy and Confidentiality

Projects involving data collection must respect privacy and confidentiality. Personal, sensitive, or confidential data must be handled securely and used only for approved academic purposes. Consent must be obtained where applicable, and data protection regulations must be followed.

Research Ethics and Human Subjects

If the FYP involves human participants, surveys, interviews, or user studies, students must ensure informed consent, voluntary participation, and confidentiality. Any potential risks to participants must be minimized.

Professional Conduct and Responsibility

Students are expected to demonstrate professionalism in interactions with supervisors, evaluators, peers, and external stakeholders. This includes respectful communication, adherence to deadlines, and responsible representation of the University.

Supervisor and Departmental Responsibilities

Supervisors are responsible for guiding students on ethical practices and monitoring compliance. The Department reserves the right to review ethical aspects of any project and take corrective action if necessary.

Handling Ethical Violations

Suspected ethical violations must be reported to the FYP Coordinator. Cases are reviewed by the Departmental Committee in accordance with university regulations. Penalties may include revision requirements, reduction of marks, project failure, or disciplinary action.

By adhering to these Ethical Guidelines, students ensure that their Final Year Project contributes positively to academic knowledge, professional practice, and societal well-being.

18. Use of AI Tools in FYP

The use of Artificial Intelligence (AI) tools in the Final Year Project (FYP) is permitted in a controlled and ethical manner. AI technologies have become an integral part of modern computing practice; therefore, students may leverage such tools to enhance productivity, learning, and problem-solving. However, the Final Year Project must fundamentally reflect the student's own intellectual contribution, technical effort, and academic understanding.

AI tools may be used to support activities such as literature search, grammar and language refinement, debugging assistance, code optimization suggestions, data visualization, and preliminary idea generation. The use of AI for these supportive purposes should enhance understanding rather than replace critical thinking or independent work.

Students are strictly prohibited from using AI tools to generate entire project reports, fabricate results, produce complete codebases without understanding, or misrepresent AI-generated content as their own original work. Any such misuse shall be treated as academic misconduct and may result in disciplinary action.

Transparency is a mandatory requirement when using AI tools. Students must clearly disclose the use of AI assistance in the Final Year Project report where applicable, particularly if AI tools significantly influenced code development, data analysis, system design, or documentation. Disclosure ensures academic honesty and protects the credibility of the work.

Supervisors are responsible for guiding students on the appropriate and ethical use of AI tools and for monitoring student work to ensure originality and genuine understanding. During viva voce examinations, students must be able to clearly explain all aspects of their project, regardless of any AI assistance used.

The Department reserves the right to review the use of AI tools in any Final Year Project. If AI usage is found to undermine academic integrity or violate university policies, appropriate penalties may be imposed in accordance with departmental and university regulations.

By adhering to these guidelines, students can responsibly integrate AI tools into their Final Year Projects while maintaining academic integrity, ethical standards, and the educational objectives of the BS Computer Science program.

19. Grievance and Appeal Mechanism

The Department of Computer Science, Abdul Wali Khan University Mardan, is committed to ensuring fairness, transparency, and due process in all matters related to the Final Year Project (FYP). The Grievance and Appeal Mechanism provides students with a formal channel to raise concerns and seek review of decisions related to supervision, evaluation, or administrative matters.

This mechanism is designed to resolve issues in a timely, impartial, and structured manner while maintaining academic integrity, professional respect, and institutional discipline.

Scope of Grievances

Students may submit grievances related to the following matters:

- Supervision-related issues, including availability or guidance concerns
- Evaluation or marking disputes
- Viva Voce conduct or procedural concerns
- Similarity or plagiarism assessment decisions
- Administrative delays or procedural irregularities

Principles Governing the Mechanism

The grievance and appeal process is governed by the principles of transparency, impartiality, confidentiality, timeliness, and fairness. All parties involved are expected to cooperate fully and maintain professional conduct throughout the process.

Informal Resolution

Students are encouraged to seek informal resolution of concerns by first discussing the matter with their supervisor or the FYP Coordinator. Many issues can be resolved amicably through clarification, guidance, or corrective action at this stage.

Formal Grievance Submission

If informal resolution is unsuccessful, students may submit a formal written grievance to the FYP Coordinator within the prescribed timeframe. The grievance must clearly state the nature of the concern, relevant details, and supporting evidence where applicable.

Review by Departmental Committee

Upon receipt of a formal grievance, the matter shall be reviewed by the Departmental FYP Committee or a designated subcommittee. The Committee may seek input from relevant parties, review documentation, and conduct hearings if required.

Decision and Communication

The decision of the Committee shall be communicated to the student in writing within a reasonable timeframe. Decisions are based on documented evidence, academic regulations, and established procedures.

Appeal Process

If a student is dissatisfied with the outcome of the grievance review, they may file an appeal to the Departmental Board of Studies (BOS) or other designated authority in accordance with university regulations. Appeals must be submitted within the stipulated period and must include clear grounds for appeal.

Final Authority

Decisions taken by the competent authority after completion of the appeal process shall be final and binding. No further appeal shall be entertained beyond the channels prescribed by the University.

Confidentiality and Record Keeping

All grievance and appeal proceedings shall be treated as confidential. Proper records of grievances, decisions, and correspondence shall be maintained by the Department for accountability and future reference.

By providing a clear and structured Grievance and Appeal Mechanism, the Department ensures that student concerns are addressed fairly while preserving the academic standards and integrity of the Final Year Project process.

20. General Regulations

The General Regulations outlined in this section govern the overall administration, conduct, and academic integrity of the Final Year Project (FYP) for the BS Computer Science program at Abdul Wali Khan University Mardan. These regulations aim to ensure transparency, consistency, fairness, and academic rigor throughout the planning, supervision, execution, and evaluation of Final Year Projects. All students, supervisors, and evaluators are required to comply with these regulations.

Only students who have successfully completed all prerequisite coursework and are officially enrolled in the final year of the BS Computer Science program are eligible to undertake the Final Year Project. Students must register their FYP in accordance with the procedures and timelines notified by the Department. Failure to complete formal registration may result in ineligibility for evaluation.

Regular attendance in supervision meetings, progress reviews, internal evaluations, and viva voce examinations is mandatory. Students are required to maintain consistent engagement with their supervisors and adhere to scheduled academic activities. Unjustified absence from any officially scheduled FYP activity may result in loss of marks or other penalties as determined by the Department.

All FYP-related activities are governed by officially approved timelines. These include proposal submission, progress reviews, internal and external evaluations, report submission, poster preparation, and viva voce examinations. Students are responsible for meeting all deadlines. Late submissions may be penalized unless prior approval has been obtained from the FYP Coordinator.

Requests for change of project title, group composition, or supervisor are discouraged once a project has been approved. Such requests may be considered only under exceptional circumstances and must be submitted in writing with valid justification. Approval rests with the FYP Coordinator and the Departmental Committee, whose decision shall be final.

Evaluation of the Final Year Project shall be conducted strictly in accordance with the approved evaluation framework. Marks awarded during evaluations are subject to finalization by the Department and shall be binding once notified. Any act of misconduct, academic dishonesty, or violation of university regulations during the FYP process may result in disciplinary action as per university statutes. The Department reserves the right to interpret and amend these regulations with the approval of competent academic authorities.

Approved by: Board of Studies (BOS)
Effective Academic Year: 2025